Aman K Gupta

Software Engineer | Backend | ML Engineer | Generative AI Engineer
+91 9335989837 | Gmail | LinkedIn | GitHub | Portfolio

Technical Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL
Backend & APIs: FastAPI, REST APIs, WebSockets, Next.js, Caching, Asynchronous Processing, Distributed Systems
Databases & Messaging: PostgreSQL, MongoDB, Redis, Firebase, Supabase, Qdrant, RabbitMQ, Celery
DevOps & Testing: AWS, GCP, Docker, Kubernetes, Git, GitHub Actions (CI/CD), Prometheus, Grafana, pytest, Linux
GenAI & LLMs: LLMs, RAG, Vector Databases, Embeddings, LangChain, LangGraph, Hugging Face Transformers, OpenAI API, Fine-tuning (LoRA/QLoRA, PEFT), Quantization, Prompt Engineering, Agentic AI
Machine Learning: PyTorch, Scikit-learn, Deep Learning, NLP, Computer Vision (OpenCV, YOLO), RNN/GRU, MLflow, MLOps
Core CS: Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Experience

Introspect Labs – Generative AI Engineer Intern | Remote Jun 2026 – Jul 2026
PersonaMatrix -Real Time Talking AI Avatar System

- Built a real-time multimodal AI service converting text/voice into emotion-aware, lip-synced video through modular STT–LLM–TTS providers using Python, FastAPI, Socket.IO, PyTorch, and MediaPipe.
- Cut per-turn latency from ∼80s to 17s (4.5x) using transport profiling, SHA-keyed caching, Opus audio, and streamed LLM output; added provider fallbacks and 55 automated tests with CI.
- Trained a context-conditioned GRU-CVAE on identity-gated footage, achieving 91% context separability; built a FACS/blendshape harness reaching 94% measured emotion recognition vs. 17% chance baseline.

DOI Soft Tech – AI Engineer Intern | Remote Jun 2025 – Aug 2025
AI Developer | Chatbot Developer

- Built and deployed supervised/unsupervised ML models, reducing data-pipeline latency by ∼40% through feature engineering and inference optimization.
- Developed an LLM-powered Python/JavaScript browser extension that automated social-media content generation, saving 10+ team hours/week.
- Deployed containerized AI services on AWS/GCP with Docker/Kubernetes; integrated REST/n8n workflows and analyzed 50K+ records using SQL, Pandas, and NumPy.

Freelance - Backend Developer | Remote Mar 2025 – Apr 2025
US-based Client

- Architected and delivered backend systems and MVPs with Next.js, Firebase, and TypeScript, applying modular system-design principles for maintainable product development.
- Streamlined backend architecture and deployment workflows to reduce deployment time by ∼30% while improving service reliability across freelance deliverables.

Projects

RAGForge | Multi-tenant RAG-as-a-Service Platform | GitHub Link | Demo Video

- Built a multi-tenant RAG-as-a-Service platform with a streaming chat API, page-level citations, an embeddable JS widget and a dashboard; hybrid search + reranker raised hit@1 from 87% to 97%.
- Built a fault-tolerant ingestion pipeline (transactional outbox, RabbitMQ retries, dead-letter queue) and a two-level Redis + semantic cache that cut repeat-answer latency from 1.3 s to 18 ms.
- Found and fixed two bottlenecks with k6 load tests: a DB connection-pool deadlock (errors 92% → 0% at 50 users) and reranker memory crashes (0 errors at 20 users; p50 2.0 s → 1.3 s).
- Deployed on AWS EC2 with Docker Compose and Caddy (HTTPS); ∼390 automated tests run in GitHub Actions CI, with Prometheus/Grafana metrics and OpenTelemetry tracing.

Distributed Task Queue & Workflow Orchestration Platform | GitHub

- Engineered a distributed task execution platform using Python, FastAPI, Redis Streams, and PostgreSQL to manage asynchronous jobs, concurrent workers, and scalable background processing.
- Designed a fault-tolerant worker architecture with heartbeat monitoring, exponential-backoff retries, idempotency, dead-letter queues, and automatic recovery from worker failures.
- Developed a DAG-based workflow orchestration engine to automate multi-stage pipelines, including AI/ML inference, data processing, and dependency-aware task execution.
- Built a real-time monitoring dashboard using Next.js, TypeScript, and WebSockets, integrating task analytics, worker health, execution logs, and Docker-based deployment.

Education

University of Petroleum and Energy Studies (UPES) 2023 – 2027
B.Tech in Computer Science Engineering - Specialization in AI & ML, Dehradun, India

Achievements & Certifications

Solved 150+ DSA problems on LeetCode, LLM Foundations | Full-Stack AI Engineer 2026 | Mastercard - Cybersecurity Job Simulation